

Igniting Global Awareness: Leveraging International Events and Agentic Engineering for Educational Hackathon Projects

I. Introduction: Setting the Stage for Innovation and Impact

The upcoming hackathon in Kent, England, presents a unique opportunity to channel the creative energies of participants towards developing innovative educational applications. Timed around late May 2025, this event can draw inspiration from several significant international observances occurring on or after May 21st. This report outlines key international events, aligns them with the United Nations Sustainable Development Goals (SDGs), and explores how Agentic Engineering techniques, specifically using Roo Coder and the create-sparc toolkit, can empower participants to build impactful applications. The core objective is to suggest ideas for projects that not only raise awareness about these global events and SDGs but also provide primary school children with tools to personalize their understanding and create relevant projects, fostering a generation of informed and engaged global citizens.

II. Spotlighting Relevant International Events (On or After May 21, 2025)

Several UN international days in late May and June 2025 offer rich thematic material for the hackathon. Each event provides a distinct lens through which participants can develop educational applications tailored for primary school children.

Table 1: Key International Events (May-June 2025)

Date	International Event	2025 Theme (Official or Anticipated)	Core Objective
May 21	World Day for Cultural Diversity for Dialogue and Development	Focus on MONDIACULT 2025 priorities & 2005 Convention goals ¹	Promote cultural diversity, intercultural dialogue for peace and sustainable development.
May 22	International Day for Biological Diversity	"Harmony with nature and sustainable development" ³	Increase understanding and awareness of biodiversity issues

			and the Kunming-Montreal Global Biodiversity Framework.
May 29	International Day of United Nations Peacekeepers	<i>Theme typically announced closer to the date</i>	Honor the contributions of UN peacekeepers and remember those who lost their lives.
May 31	World No Tobacco Day	<i>Theme typically announced closer to the date</i>	Highlight health risks associated with tobacco use and advocate for effective policies.
June 5	World Environment Day	"#BeatPlasticPollution" ⁵	Encourage awareness and action for the protection of the environment, focusing on plastic pollution.
June 8	World Oceans Day	"Wonder: Sustaining What Sustains Us" ⁷ (UN 2024: "Awaken New Depths" ⁸)	Inform public of human impacts on the ocean, mobilize sustainable ocean management.
June 12	World Day Against Child Labour	<i>Theme typically announced closer to the date</i>	Raise awareness and activism to prevent child labour.
June 20	World Refugee Day	UN theme TBD; 2024: "For a World Where Refugees Are Welcomed" ⁹	Honor refugees, build empathy, and recognize their resilience.

For the purpose of this hackathon, focusing on themes with strong educational potential for primary school children, the following events are particularly salient:

A. World Day for Cultural Diversity for Dialogue and Development (May 21)

Celebrated annually on May 21st, this day was established by the UN General Assembly in 2002 following UNESCO's adoption of the Universal Declaration on Cultural Diversity in 2001.¹⁰ Its core purpose is to highlight the richness of the world's cultures and the crucial role of intercultural dialogue for achieving peace and sustainable development.² The UN notes that bridging cultural gaps is essential for peace, stability, and development, and that three-quarters of the world's major conflicts have a cultural dimension.¹¹

The 2025 observance will likely reflect the priorities emerging from the MONDIACULT 2022 conference, which affirmed culture as a "global public good" and called for its integration into the post-2030 development agenda.¹ The upcoming MONDIACULT 2025 conference in Barcelona (September 29 - October 1, 2025) will further assess these priorities.¹ The 2025 theme for Dialogue and Diversity Day, as suggested by some sources, centers on achieving the four goals of the 2005 UNESCO Convention on the Protection and Promotion of the Diversity of Cultural Expressions: supporting sustainable systems of governance for culture, achieving a balanced flow of cultural goods and services, increasing mobility of artists and cultural professionals, and integrating culture in sustainable development frameworks.²

For primary school children, the complex language of international conventions and global public goods needs careful translation. The essence of MONDIACULT's ambition – valuing every culture and ensuring culture helps build a better future – can be distilled into age-appropriate concepts. Hackathon projects can focus on celebrating cultural expressions (stories, music, art, traditions) and fostering curiosity about different ways of life. An application might allow children to virtually visit different countries, learn simple greetings, explore traditional crafts, or listen to folk tales. This approach directly supports the educational aim of personalizing understanding, making abstract global concepts tangible and engaging for young learners.

B. International Day for Biological Diversity (May 22)

Observed annually on May 22nd, this day aims to increase understanding and awareness of biodiversity issues.¹³ The UN designated this day to highlight the significant challenges human activities pose to biological diversity and to recognize biodiversity as a global asset crucial for future generations.¹³

The official theme for 2025 is **“Harmony with nature and sustainable development”**.³ This theme directly connects the Kunming-Montreal Global Biodiversity Framework (also known as The Biodiversity Plan) with the Sustainable Development Goals, emphasizing that these agendas are mutually supportive and

must advance together.³ The Biodiversity Plan, adopted in December 2022, sets ambitious targets, including restoring 30% of degraded ecosystems by 2030 and significantly reducing the introduction of invasive alien species.³ The 2025 campaign aims to instill a sense of urgency, as only five years will remain to meet near-term biodiversity targets and the SDGs.³

For primary school children in Kent, this theme offers a chance to explore local ecosystems. "Harmony with nature" can be interpreted through apps that encourage children to identify local plants and animals, understand their roles, and learn about conservation efforts. The concept of interconnectedness within The Biodiversity Plan – how protecting one species or habitat benefits others – is a valuable lesson. An app could, for instance, simulate the effects of reintroducing a native species to a local environment or show how reducing pollution helps wildlife thrive. This fosters an early appreciation for the natural world and humanity's role in its stewardship, making the global goals of biodiversity conservation personally relevant.

C. World Environment Day (June 5)

Celebrated annually on June 5th, World Environment Day is the UN's principal vehicle for encouraging worldwide awareness and action for the environment.¹⁶ Led by the UN Environment Programme (UNEP) since 1973, it has grown into the largest global platform for environmental outreach.¹⁶

The theme for World Environment Day 2025 is "**#BeatPlasticPollution**", and the global celebrations will be hosted by the Republic of Korea.⁵ This theme aims to mobilize communities to implement and advocate for solutions to the plastic pollution crisis, spotlighting scientific evidence on its impacts and driving momentum to refuse, reduce, reuse, recycle, and rethink plastic use.⁵ This reinforces the global commitment made in 2022 to develop a legally binding global instrument to end plastic pollution.⁶ More than 400 million tonnes of plastic are produced annually, with half designed for single-use, and less than 10% recycled.⁵

The "Beat Plastic Pollution" theme is highly concrete and actionable for primary school children. Projects can focus on identifying different types of plastic, understanding their lifecycle, learning about recycling processes in Kent, and brainstorming creative ways to reduce plastic use at home and school. The host country, the Republic of Korea, provides excellent inspiration. Their "full life-cycle plastic strategy" and specific initiatives like Jeju Province's vision to be plastic-free by 2040, along with its advanced waste separation system⁶, can be adapted into educational game mechanics. For instance, an app could simulate Jeju's waste sorting system, teaching children in Kent the importance of proper disposal and the impact of

their choices. This approach connects a global event to local action through the host country's example, making the theme tangible and showcasing real-world solutions.

D. World Oceans Day (June 8)

Observed annually on June 8th, World Oceans Day provides an opportunity to honor, help protect, and conserve the ocean.¹⁸ The UN marks this day to celebrate the ocean's importance and to foster a better understanding of how to interact with it sustainably.¹⁸ The ocean produces at least 50% of the planet's oxygen and absorbs about 30% of carbon dioxide produced by humans.⁷

While the official UN page⁸ listed the 2024 theme as "Awaken New Depths," other sources suggest the 2025 theme will be **"Wonder: Sustaining What Sustains Us"**.⁷ This theme aims to highlight the awe-inspiring nature of the ocean and our collective responsibility to protect it.⁷ This is particularly poignant given that, despite our total dependence on the ocean, only about 10% of it has been explored.⁷

The theme of "Wonder" is a powerful motivator for learning, especially for primary school children. The mystery and unexplored nature of the oceans can be a springboard for apps that encourage virtual exploration, such as simulated submersible trips to discover unique marine species, or interactive storytelling about ocean myths and real-life oceanographic expeditions. Even for children in Kent, who may not live directly on the coast, connections can be made by exploring how local rivers eventually lead to the sea and how actions inland impact marine environments. This theme can ignite curiosity and a passion for scientific inquiry, making learning about ocean conservation an adventure.

E. World Refugee Day (June 20)

Celebrated annually on June 20th, World Refugee Day was established by the UN in 2001 to honor refugees around the world, coinciding with the 50th anniversary of the 1951 Convention Relating to the Status of Refugees.¹⁹ The day aims to build empathy and understanding for the plight of refugees and to recognize their strength and resilience in rebuilding their lives.⁹

The official UN theme for 2025 is yet to be announced in the provided materials; the 2024 theme was "For a World Where Refugees Are Welcomed".⁹ Other related themes for 2025 include Refugee Week Australia's "Finding Freedom: Diversity in Community"²¹ and Pope Francis's chosen theme for the Catholic Church's observance, "Migrants are missionaries of hope".²²

For primary school children, themes focusing on "welcome" and "diversity in

community" are more accessible and positive than dwelling directly on the hardships of displacement. Hackathon projects can revolve around creating virtual welcoming spaces, celebrating the contributions of diverse community members (including those with refugee backgrounds), and sharing stories of friendship that transcend cultural differences. An app could allow children to design a welcoming school or neighborhood, or feature interactive stories about children from different parts of the world becoming friends and learning from each other. This approach fosters empathy and an appreciation for diversity in a child-friendly manner.

III. Connecting to Global Priorities: The UN Sustainable Development Goals (SDGs)

The 2030 Agenda for Sustainable Development, adopted by all UN member states in 2015, provides a shared blueprint for peace and prosperity for people and the planet, now and into the future.²³ At its heart are the 17 Sustainable Development Goals (SDGs), which are an urgent call for action by all countries – developed and developing – in a global partnership. They recognize that ending poverty and other deprivations must go hand-in-hand with strategies that improve health and education, reduce inequality, and spur economic growth – all while tackling climate change and working to preserve our oceans and forests.²³ The SDGs are integrated, acknowledging that action in one area will affect outcomes in others, and that development must balance social, economic, and environmental sustainability.²⁴

Identifying Relevant SDGs for the Hackathon

Given the hackathon's educational theme and the focus on the aforementioned international events, several SDGs are particularly relevant. Applications developed can help primary school children understand these goals in an accessible way:

- **SDG 4: Quality Education:** Aims to "ensure inclusive and equitable quality education and promote lifelong learning opportunities for all".²³ This is the foundational SDG for the hackathon.
 - *Target 4.7* is especially pertinent: "By 2030, ensure that all learners acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship and appreciation of cultural diversity and of culture's contribution to sustainable development".²³ Educational apps can directly address this by teaching about SDGs, cultural diversity, and environmental stewardship.
- **SDG 10: Reduced Inequalities:** Aims to "reduce inequality within and among

countries".²³

- *Target 10.2*: "By 2030, empower and promote the social, economic and political inclusion of all, irrespective of age, sex, disability, race, ethnicity, origin, religion or economic or other status".²³ This connects to themes from World Day for Cultural Diversity and World Refugee Day, encouraging apps that promote understanding and inclusion.
- **SDG 12: Responsible Consumption and Production**: Aims to "ensure sustainable consumption and production patterns".²³
 - *Target 12.5*: "By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse".²³ This directly links to the World Environment Day theme of #BeatPlasticPollution.
 - *Target 12.8*: "By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature".²³ Educational apps are a prime vehicle for this.
- **SDG 13: Climate Action**: Aims to "take urgent action to combat climate change and its impacts".²³
 - *Target 13.3*: "Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning".²³ This is highly relevant for apps addressing environmental themes.
- **SDG 14: Life Below Water**: Aims to "conserve and sustainably use the oceans, seas and marine resources for sustainable development".²³
 - *Target 14.1*: "By 2025, prevent and significantly reduce marine pollution of all kinds, in particular from land-based activities, including marine debris and nutrient pollution".²³ This aligns with World Oceans Day and the plastic pollution theme.
- **SDG 15: Life on Land**: Aims to "protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss".²³
 - *Target 15.5*: "Take urgent and significant action to reduce the degradation of natural habitats, halt the loss of biodiversity and, by 2020, protect and prevent the extinction of threatened species".²³ This is central to the International Day for Biological Diversity.

Strategies for Making SDGs Accessible to Primary School Children

Translating the complexities of the SDGs into content suitable for primary school children requires creativity and pedagogical thoughtfulness. Key strategies include:

- **Storytelling**: Narratives featuring relatable characters facing challenges and

finding solutions related to SDGs can be highly engaging.

- **Gamification:** Turning learning about SDGs into games with clear objectives, challenges, and rewards can enhance motivation and understanding.
- **Visuals and Interactivity:** Using bright, child-friendly graphics, animations, and interactive elements can make abstract concepts more concrete.
- **Local Relevance:** Connecting global goals to the child's immediate environment (e.g., how reducing waste at their school helps the planet) makes the SDGs more tangible.
- **Focus on Positive Action:** Emphasizing what children *can do* empowers them and prevents feelings of being overwhelmed. Resources like the UN's "SDG Book Club" offer examples of age-appropriate literature for each goal.²³

The inherent interconnectedness of the SDGs provides a powerful narrative framework. For primary school children, who are adept at understanding simple cause-and-effect relationships, this can be a compelling teaching tool. An application focusing on reducing plastic pollution (SDG 12, SDG 14) can illustrate how this action also contributes to protecting marine animals (SDG 15) and ensuring cleaner water, which benefits human health (SDG 3). This approach not only makes the learning experience richer but also subtly introduces systems thinking, reflecting the real-world complexity of achieving sustainable development.

IV. Empowering Innovation: Agentic Engineering with Roo Coder and create-sparc

The hackathon participants will be utilizing Agentic Engineering techniques, specifically through the create-sparc toolkit and the Roo Coder VS Code extension. These tools offer powerful capabilities to design and build the educational applications envisioned.

Overview of Agentic Engineering Principles

Agentic Engineering involves designing and building software agents—autonomous or semi-autonomous systems—that can perceive their environment (or user inputs), reason about their goals, make decisions, and take actions to achieve those goals. In the context of this hackathon, the "agents" will be the educational applications themselves. These apps will be designed to guide primary school children through interactive learning experiences, respond to their inputs, and help them achieve specific educational objectives related to international events and SDGs.

Key Features and Capabilities of create-sparc and Roo Coder

- **create-sparc Toolkit:** This toolkit is designed to streamline and enhance software development by integrating AI assistance and structured methodologies.²⁶
 - **SPARC Methodology:** Stands for Specification, Pseudocode, Architecture, Refinement, and Completion.²⁶ It provides a structured framework for deconstructing complex projects into manageable subtasks, ensuring clarity and quality throughout the development lifecycle.²⁶ This is invaluable for hackathon teams to organize their work efficiently under time pressure.
 - **AI-Driven Code Generation (AIGI):** This framework within create-sparc leverages AI for code generation, incorporating prompt engineering, code quality assessment, reflection, iterative improvement, and context management to produce optimal code outputs.²⁶ This can significantly accelerate the development process.
 - **Model Context Protocol (MCP):** MCP enables AI agents to securely connect with external services such as databases, APIs, and cloud resources.²⁶ It prioritizes security through environment variable references for credentials and permission scoping, following the principle of least privilege.²⁶ This is crucial for applications that might need to fetch real-time data or integrate with other educational platforms. The MCP Wizard helps manage configurations for these connections.²⁶
 - **Boomerang Task Concept:** This integrates the SPARC methodology with Roo Code, allowing development tasks to "bounce" through specialized AI agents, each handling a specific phase from specification to completion, thereby maintaining structure and efficiency.²⁶
 - **Component Generator:** The command `npx create-sparc add component --name MyComponent` facilitates modular development, allowing teams to easily add new, self-contained features or sections to their applications.²⁷
- **Roo Coder (VS Code Extension):** Roo Coder is an open-source AI-powered autonomous coding agent that integrates directly into the Visual Studio Code editor.³⁰
 - **AI-Powered Coding Assistance:** It can generate, refactor, debug, and document code across multiple languages based on natural language prompts from the developer.³⁰
 - **Specialized Modes:** Roo Coder offers various modes like 'Code', 'Architect', 'Debug', and 'Ask', along with the ability to create custom modes. These specialized agents can handle different aspects of development, from high-level design to detailed implementation and troubleshooting.³⁰
 - **File Manipulation and Terminal Execution:** The agent can read and write files directly in the editor and execute terminal commands, enabling

automation of various development and deployment tasks.³⁰

- **Flexible Model Integration:** It supports OpenAI-compatible models and allows integration with custom APIs, providing flexibility in choosing the underlying AI capabilities.³⁰

How These Tools Facilitate Educational Applications for Primary School Children

The combination of create-sparc and Roo Coder offers significant advantages for developing the types of educational applications envisioned for this hackathon:

- **Rapid Prototyping:** Teams can quickly create functional prototypes of interactive stories, simple games, or quiz modules. Roo Coder can generate boilerplate code, initial content, or UI elements based on descriptive prompts, allowing for fast iteration.
- **Personalized Learning Paths:** Agentic applications can be designed to adapt content or difficulty based on a child's interactions and responses. The SPARC methodology can help structure these adaptive pathways, and Roo Coder can assist in implementing the logic.
- **Content Generation Support:** Roo Coder can be prompted to help draft age-appropriate explanations of SDG concepts, create dialogues for characters in educational stories, or generate sets of quiz questions aligned with learning objectives.
- **Interactive Narratives:** "Choose your own adventure" style stories related to event themes (e.g., a child making choices to help reduce plastic pollution) can be developed, with Roo Coder helping to script the various narrative branches and outcomes.
- **Project-Based Learning Modules:** The user query specifically mentions children using the apps "to create relevant projects." The hackathon applications can act as guide-agents, leading children through the process of planning and executing simple projects (e.g., designing a poster about local wildlife, or planning a week of plastic-free lunches).

The SPARC methodology (Specification, Pseudocode, Architecture, Refinement, Completion), while designed for software development²⁶, offers a surprisingly effective pedagogical framework that can be embedded within the educational apps themselves. Children engaging with an app to "create relevant projects" can be guided through a simplified, child-friendly version of SPARC. For instance, the stages could be presented as: "1. What's Your Big Idea? (Specification)," "2. Plan Your Steps (Pseudocode)," "3. Gather Your Tools & Build It! (Architecture)," "4. Make It Even Better! (Refinement)," and "5. Show the World! (Completion)." By using tools built with SPARC to teach a simplified SPARC process, the hackathon can deliver a meta-level

educational benefit, equipping children with fundamental project management and problem-solving skills applicable far beyond the app itself.

V. Hackathon Project Blueprints: Ideas for Impact

This section provides five detailed project blueprints, each aligning an international event with educational objectives for primary school children, relevant SDGs, and specific ways to leverage Agentic Engineering tools.

1. Project Title: "Culture Explorer Kids"

- **Target International Day & Theme Alignment:** World Day for Cultural Diversity for Dialogue and Development (May 21). Aligns with promoting intercultural dialogue and understanding the value of diverse cultures.¹
- **Primary School Educational Objectives:**
 - Children will learn basic facts (e.g., greetings, common foods, a traditional art form or story, a significant festival) about 3-5 different cultures.
 - Develop curiosity, respect, and appreciation for cultural differences.
 - Understand that diversity is a strength.
- **Relevant SDG(s):**
 - **Primary:** SDG 4 (Quality Education - Target 4.7: appreciation of cultural diversity).
 - **Secondary:** SDG 10 (Reduced Inequalities - Target 10.2: promote social inclusion).
- **Application Features & Functionality (The "Agentic" App):**
 - An interactive, animated world map. Children can tap on different countries/regions.
 - Tapping a region introduces a child character (an avatar) from that culture.
 - The character shares simple, voice-narrated facts, teaches a common greeting (audio and text), shows images or simple animations of traditional food, clothing, or festivals.
 - A mini-game or interactive story related to that culture (e.g., helping prepare for a festival, learning a simple craft step-by-step).
 - "My Culture Passport" feature: Children can collect "stamps" for each culture explored and draw or write something they learned or liked, fostering reflection and project creation.
- **Leveraging Roo Coder/create-sparc:**
 - **Roo Coder (Natural Language to Code/Content):**
 - Generate initial drafts for short, age-appropriate stories or factual snippets for each culture based on prompts like: "Write a simple 100-word

introduction to Japanese culture for 6-year-olds, mentioning cherry blossoms, sushi, and a simple greeting."

- Assist in creating dialogue scripts for the cultural avatars.
- **create-sparc (SPARC Methodology):**
 - Use 'Specification' to define learning objectives and content points for each cultural module.
 - Use 'Architecture' to plan the navigation flow between the map and individual culture screens.
- **create-sparc (AIGI):**
 - Refine AI-generated text for tone, simplicity, and accuracy suitable for primary children.
- **create-sparc (Component Generator):** Create a reusable "CultureModule" component that can be populated with different content for each featured culture.
- **Potential for Awareness/Contribution:** Promotes intercultural understanding, empathy, and global citizenship from a young age, directly contributing to the spirit of Dialogue and Diversity Day.

2. Project Title: "Kent Wildlife Protectors"

- **Target International Day & Theme Alignment:** International Day for Biological Diversity (May 22). Theme: "Harmony with nature and sustainable development".³
- **Primary School Educational Objectives:**
 - Identify 5-7 common local wildlife species (animals and plants) in Kent.
 - Understand basic needs of these species and simple threats they face (e.g., litter, habitat loss).
 - Learn about positive actions to help local wildlife (e.g., creating a small bug hotel, planting bee-friendly flowers, responsible waste disposal).
- **Relevant SDG(s):**
 - **Primary:** SDG 15 (Life on Land - Target 15.5: halt biodiversity loss, protect threatened species).
 - **Secondary:** SDG 4 (Quality Education - Target 4.7: education for sustainable development), SDG 11 (Sustainable Cities and Communities - local environment focus).
- **Application Features & Functionality (The "Agentic" App):**
 - "My Kent Safari": An interactive map of a generic Kent park/garden. Children can tap on hotspots to discover animated local animals/plants.
 - Each species has a simple profile: picture, sound, fun fact, what it eats, where it lives.
 - Mini-games: e.g., "Feed the Bird" (matching correct food), "Clean the Park"

- (sorting litter from natural items).
 - "Wildlife Protector Pledge": Children can choose simple actions to help wildlife.
 - "My Wildlife Project": Guides children to create a simple project, like drawing their favorite local animal and writing why it's important, or designing a poster about keeping parks clean for animals.
- **Leveraging Roo Coder/create-sparc:**
 - **Roo Coder (Natural Language to Code/Content):**
 - Generate Python/JavaScript snippets for mini-game logic (e.g., "Create a drag-and-drop game where users sort items into 'Recyclable', 'Compost', 'Landfill' bins").
 - Draft simple descriptive texts for local species based on factual prompts.
 - **create-sparc (MCP - Model Context Protocol):**
 - Potentially connect to an API (if a child-safe one exists) for images or verified fun facts about UK wildlife, ensuring data is fetched securely.
 - **create-sparc (SPARC - Refinement):**
 - Use AI assistance to review and simplify game instructions and species information for clarity and age-appropriateness.
- **Potential for Awareness/Contribution:** Raises awareness of local biodiversity and empowers children with actionable steps to contribute to its conservation, embodying "Harmony with nature."

3. Project Title: "Plastic Patrol Challenge"

- **Target International Day & Theme Alignment:** World Environment Day (June 5). Theme: "#BeatPlasticPollution".⁵
- **Primary School Educational Objectives:**
 - Understand that plastic pollution is harmful to the environment and animals.
 - Identify common single-use plastic items.
 - Learn about the 3 R's (Reduce, Reuse, Recycle) with a focus on plastics.
 - Be inspired by real-world solutions (like those in host country South Korea ⁶).
- **Relevant SDG(s):**
 - **Primary:** SDG 12 (Responsible Consumption and Production - Target 12.5: reduce waste generation; Target 12.8: information for sustainable lifestyles).
 - **Secondary:** SDG 14 (Life Below Water - Target 14.1: reduce marine pollution), SDG 15 (Life on Land).
- **Application Features & Functionality (The "Agentic" App):**
 - "Plastic Detective": An interactive game where children identify plastic items in different scenes (home, park, beach) and learn if they can be recycled or how they could be reduced/reused.

- "Recycle Robot Sorter": A fast-paced game simulating a recycling plant (inspired by systems like Jeju's ⁶), where children sort plastics correctly.
- "My Eco-Invention Lab": Children can draw or describe ideas for new uses for old plastic items or alternatives to plastic products. This serves as their "relevant project."
- Short animated stories showing the impact of plastic on animals and how reducing plastic helps.
- **Leveraging Roo Coder/create-sparc:**
 - **Roo Coder (Specialized Modes):**
 - Use 'Code' mode to generate game logic for the "Recycle Robot Sorter."
 - Use 'Ask' mode to get ideas for simplifying concepts about plastic lifecycles for young children.
 - **create-sparc (AIGI - AI-Driven Code Generation):**
 - Generate visual assets or descriptions for different plastic types and their recycled products, which can then be refined.
 - Iteratively improve game difficulty based on defined learning progression.
 - **create-sparc (SPARC - Pseudocode):**
 - Clearly outline the logic for the "Plastic Detective" game, detailing how different items are categorized and what feedback the child receives.
- **Potential for Awareness/Contribution:** Directly addresses the #BeatPlasticPollution theme, making children aware of the problem and empowering them with knowledge and ideas for action.

4. Project Title: "Ocean Wonders Explorer"

- **Target International Day & Theme Alignment:** World Oceans Day (June 8). Anticipated Theme: "Wonder: Sustaining What Sustains Us".⁷
- **Primary School Educational Objectives:**
 - Develop a sense of wonder and curiosity about the ocean and marine life.
 - Learn about 3-5 fascinating marine creatures and their habitats.
 - Understand that human actions (like pollution) can affect the ocean, even from far away (e.g., rivers in Kent leading to the sea).
 - Explore the concept of ocean conservation in simple terms.
- **Relevant SDG(s):**
 - **Primary:** SDG 14 (Life Below Water - Target 14.1: reduce marine pollution; Target 14.2: protect marine ecosystems).
 - **Secondary:** SDG 4 (Quality Education), SDG 13 (Climate Action - oceans' role in climate).
- **Application Features & Functionality (The "Agentic" App):**
 - "Virtual Submersible Adventure": Children navigate a simple animated

- submersible through different ocean zones (e.g., coral reef, deep sea).
 - Discover and collect "Wonder Cards" for different marine animals, featuring beautiful visuals, sounds, and a "Did You Know?" amazing fact.
 - "River to Ocean Journey": An interactive animation showing how a piece of litter dropped in a Kent river could travel to the ocean, highlighting interconnectedness.
 - "My Ocean Promise": Children can choose a simple promise (e.g., "I will not drop litter," "I will learn more about fish") and create a drawing of their favorite sea creature to accompany it.
- **Leveraging Roo Coder/create-sparc:**
 - **Roo Coder (Natural Language to Content):**
 - Generate engaging "Did You Know?" facts about marine species suitable for 6-8 year olds.
 - Draft scripts for the voice-over in the "Virtual Submersible Adventure."
 - **create-sparc (SPARC - Architecture):**
 - Design the different levels or zones for the submersible adventure, outlining key discoveries in each.
 - **create-sparc (MCP):**
 - If a suitable, curated API exists for child-friendly ocean facts or images (e.g., from a marine conservation society), MCP could be used to fetch this content.
 - **Roo Coder (Custom Modes):**
 - A custom mode could be configured to help generate story prompts for children to write their own short ocean adventures, guided by the app.
- **Potential for Awareness/Contribution:** Instills a love and respect for the ocean, connecting local actions to global marine health, and fostering the "wonder" that can lead to lifelong engagement.

5. Project Title: "Welcome Friends: Our Diverse Community"

- **Target International Day & Theme Alignment:** World Refugee Day (June 20). Aligns with themes like "For a World Where Refugees Are Welcomed" ⁹ and "Finding Freedom: Diversity in Community". ²¹
- **Primary School Educational Objectives:**
 - Develop empathy and understanding for people from different backgrounds, including those who may have had to leave their homes.
 - Learn that everyone deserves to feel welcome and safe.
 - Celebrate the positive aspects of diversity in a community.
 - Understand simple ways to be a good friend to someone new.
- **Relevant SDG(s):**

- **Primary:** SDG 10 (Reduced Inequalities - Target 10.2: promote social inclusion; Target 10.7: facilitate orderly, safe, regular and responsible migration and mobility of people).
- **Secondary:** SDG 4 (Quality Education - Target 4.7: global citizenship, promotion of a culture of peace), SDG 16 (Peace, Justice and Strong Institutions - fostering inclusive societies).
- **Application Features & Functionality (The "Agentic" App):**
 - Interactive stories about children from different backgrounds becoming friends and helping each other.
 - "Design a Welcome Poster/Card": A creative tool allowing children to design a welcoming message for a new student or neighbor. This is their "relevant project."
 - "Our Friendship Quilt": Children can contribute a virtual "patch" with a drawing or word representing friendship or welcome, creating a collaborative digital quilt.
 - Simple scenarios/dilemmas: "What would you do if a new child in class looks lonely?" with positive choices offered.
- **Leveraging Roo Coder/create-sparc:**
 - **Roo Coder (Natural Language to Story/Scenario):**
 - Generate drafts for the interactive stories, focusing on themes of empathy, kindness, and inclusion. E.g., "Write a short story for 7-year-olds about two children from different countries who become friends at a new school."
 - Create simple scenario prompts for the dilemma-based interactions.
 - **create-sparc (SPARC - Specification & Refinement):**
 - Clearly define the emotional and learning objectives for each story and scenario.
 - Use AI assistance to refine the language in stories to ensure it is sensitive, positive, and age-appropriate.
 - **Roo Coder (Agent Specializations):**
 - Potentially use a "Product Manager" or "QA" like mode (conceptual use of Roo Coder's capabilities) to review content for appropriateness and positive messaging.
- **Potential for Awareness/Contribution:** Fosters empathy, reduces prejudice, and promotes a welcoming attitude towards newcomers, reflecting the core spirit of World Refugee Day in a way that is positive and empowering for young children.

Table 2: Hackathon Project Idea Matrix (Summary)

International Event	Project Idea Title	Target SDG(s)	Brief App Description	Key Agentic Engineering Application	Core Educational Goal for Children
World Day for Cultural Diversity for Dialogue & Development	Culture Explorer Kids	SDG 4, SDG 10	Interactive map and cultural avatars teach about global cultures; "Culture Passport" creation.	Roo Coder for story/fact generation; create-sparc for module structure.	Develop curiosity and respect for cultural differences.
International Day for Biological Diversity	Kent Wildlife Protectors	SDG 15, SDG 4, SDG 11	Discover local Kent wildlife, play conservation mini-games, and create a "Wildlife Project."	Roo Coder for game logic snippets; MCP for wildlife data API (optional).	Identify local species and learn simple conservation actions.
World Environment Day	Plastic Patrol Challenge	SDG 12, SDG 14, SDG 15	Identify plastics, learn 3 R's through games, design eco-inventions; inspired by Korean waste systems.	Roo Coder for game logic; AIGI for asset generation & game refinement.	Understand plastic pollution and ways to reduce/reuse/recycle.
World Oceans Day	Ocean Wonders Explorer	SDG 14, SDG 4, SDG 13	Virtual submersible adventure to discover marine life; learn how local actions impact	Roo Coder for fact/script generation; create-sparc for adventure design.	Develop wonder for oceans and understand conservation basics.

			oceans; "Ocean Promise" creation.		
World Refugee Day	Welcome Friends: Our Diverse Community	SDG 10, SDG 4, SDG 16	Interactive stories on friendship & inclusion; design "Welcome Posters"; contribute to a digital "Friendship Quilt."	Roo Coder for story/scenario generation; create-sparc for refinement.	Develop empathy and celebrate diversity in communities .

VI. Conclusion: Fostering a Generation of Changemakers

The confluence of these significant international events in late May and June 2025, coupled with the robust framework of the UN Sustainable Development Goals, provides a fertile ground for the Kent hackathon. By harnessing the capabilities of Agentic Engineering through Roo Coder and the create-sparc toolkit, participants are well-equipped to develop truly innovative and engaging educational applications for primary school children.

The project blueprints outlined offer tangible pathways to translate global themes into personalized, actionable learning experiences. These applications can empower children not merely as passive recipients of information, but as active creators and thinkers, capable of understanding complex issues like cultural diversity, biodiversity loss, plastic pollution, ocean conservation, and the importance of welcoming communities. The emphasis on enabling children to "create relevant projects" using these apps further deepens their engagement and fosters a sense of agency.

The hackathon itself, by bringing together diverse talents to work towards common goals with a social purpose, mirrors the spirit of SDG 17: Partnerships for the Goals.²³ This collaborative endeavor, focused on leveraging technology for education and awareness, can be a powerful demonstration of how multi-stakeholder efforts can drive positive change.²³

Ultimately, the impact of this hackathon extends beyond the event itself. By fostering

awareness, empathy, critical thinking, and problem-solving skills from a young age, these technology-driven educational projects can play a vital role in nurturing a generation of informed, responsible, and proactive global citizens – the very changemakers who will shape a more sustainable and equitable future.

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